Module 01: Systems -- The Interconnected Web of Your Life

Learning Objectives

1. Define a **system** as a set of interacting components separated from its environment by a boundary, and identify systems in everyday life.
2. Explain the concept of a **Markov Blanket** as the formal boundary that separates a system's internal states from its external environment.
3. Analyze how your social world -- family, school, friend groups, online communities -- consists of nested and overlapping systems.
4. Recognize that systems maintain their identity not through fixed parts but through persistent patterns of interaction.

Introduction

Think about the different "worlds" you move through in a single day. You wake up in your family system -- a set of relationships, routines, and unspoken rules. You enter the school system, with its own norms, hierarchies, and expectations. You check your phone and step into a digital social system that operates by yet another set of dynamics. Each of these is a **system**: a collection of parts that interact in organized ways and that can be meaningfully distinguished from everything outside them.

But what makes these systems *systems* rather than random collections of people? Active Inference provides a precise answer: a system is anything that maintains a **Markov Blanket** -- a statistical boundary that separates what is inside from what is outside, while allowing controlled interactions between the two.

Key Concepts

1. What Makes Something a System?

A pile of sand is not a system. A family is. The difference is **organization** -- the parts of a system interact with each other in ways that create patterns, and those patterns persist over time. Your friend group has a recognizable character that survives individual members joining or leaving. Your school has an identity that persists even as every student eventually graduates. Systems are defined by their relationships, not their parts.

2. Markov Blankets in Social Life

In Active Inference, a system's boundary is formalized as a **Markov Blanket**: a set of states that mediates all interactions between inside and outside. In your social life, think of the boundary of your friend group. Information from outside (gossip, new people, cultural trends) reaches the group through specific channels -- particular members who bridge different social circles. Information from inside (inside jokes, group decisions, shared memories) reaches the outside world through those same bridges. The Markov Blanket is not a wall; it is a *filter*.

3. Nested and Overlapping Systems

Your life is not one system but many, layered and overlapping. You are a system (with your skin as a physical Markov Blanket). Your family is a system that contains you. Your school is a system that contains your family's interactions with it. And you belong to multiple systems simultaneously -- your sports team, your online communities, your classroom. Active Inference handles this through the concept of **hierarchical generative models**: systems within systems, each maintaining its own blanket at its own level.

4. Systems Resist Dissolution

A key insight from Active Inference is that any system that continues to exist must be actively maintaining its boundary. A friend group that stops communicating dissolves. A family that stops interacting fragments. The maintenance of a system's Markov Blanket is not passive -- it requires ongoing effort, which Active Inference formalizes as the minimization of **variational**

free energy. To exist as a system is to continuously work against the tendency toward disorder.

5. Boundaries Can Shift

System boundaries are not fixed. When you start dating someone, the Markov Blanket of your social system expands to include them -- information flows differently, new sensory states become relevant. When a friendship ends, the boundary contracts. Recognizing that boundaries are dynamic helps explain why transitions in your social life (changing schools, entering college, losing a friend) feel so disorienting: your system's blanket is being restructured.

Active Inference Connection

The free energy principle states that any self-organizing system must minimize the difference between its predictions and its actual sensory states. For the social systems in your life, this means that groups tend to develop shared expectations, norms, and routines that reduce surprise for their members. When someone violates a group norm, the resulting prediction error creates tension -- the group must either update its model (accommodate the change) or act to restore the expected pattern (enforce the norm).

Applications

- **Case Study 1 -- The Transfer Student**: When a new student joins a tight-knit class, both the student and the class experience elevated prediction error. The new student's generative model of social dynamics does not match this particular group. The class's model does not include this person. Over weeks, mutual model-updating occurs: the boundary of the class system expands, internal dynamics reorganize, and a new equilibrium emerges. This process *is* Active Inference operating at the social level.
- **Case Study 2 -- Family Systems During a Move**: When a family relocates to a new city, the family system's external environment changes dramatically. Routines break, familiar landmarks disappear, social connections sever. The family must minimize free energy by building new routines, forming new connections, and updating its collective

model of "how things work." Families that successfully adapt are, in Active Inference terms, effectively minimizing prediction error in a novel environment.

Discussion Questions

1. Identify three systems you belong to. For each, what serves as the Markov Blanket -- what mediates the flow of information between inside and outside?
2. Why do you think it feels uncomfortable to be in a social situation where you do not know the unspoken rules? How does Active Inference explain this discomfort?
3. Can a social media group chat be considered a system in the Active Inference sense? What is its boundary, and how is it maintained?

Summary

Your everyday life is structured by systems -- family, school, friendships, online communities -- each defined by a boundary that separates inside from outside while allowing controlled interaction. Active Inference formalizes this through the Markov Blanket, showing that system boundaries are not arbitrary but are actively maintained through the ongoing minimization of free energy. In Module 02, we move from the system to what is inside it: the **agent** -- you -- and the generative models that define who you are.